

Flight Tracking & Operations Dashboard

Design Explanation

Introduction

For this assignment, I designed and developed a Flight Tracking & Operations Dashboard intended for aviation operations teams. The goal was to create a clean and user-friendly interface that allows users to quickly monitor flight activity, view operational metrics, and access detailed flight information from a single dashboard.

While working on the design, I focused on creating a layout that is simple to understand, visually organized, and easy to use. Since aviation personnel may need to monitor multiple flights at once, I wanted the most important information to be visible immediately without requiring unnecessary navigation.

Dashboard Layout

The dashboard follows a structured layout where operational information is divided into clear sections.

At the top of the page, KPI cards display key flight statistics such as Total Flights, Active Flights, Delayed Flights, and Arrived Flights. These metrics provide a quick overview of the current operational status and help users understand the situation at a glance.

Below the KPI section, the interactive flight map becomes the primary focus of the screen. Since the assignment is centered around flight tracking, I intentionally gave the map the largest portion of the available space. This allows users to focus on flight locations and routes without distractions.

The dashboard also includes search and filtering options, making it easier to locate specific flights based on callsign, status, origin airport, or destination airport.

Flight Details Experience

One of the key requirements was displaying detailed flight information when a flight is selected.

To achieve this, I designed a dedicated flight details panel that appears alongside the map. This panel displays important flight information such as:

- Flight Number
- Callsign
- Aircraft Type
- Origin Airport
- Destination Airport
- Current Status
- Estimated Departure Time
- Estimated Arrival Time

Keeping this information in a separate panel allows users to view detailed flight data without losing context of the map or dashboard.

Route Visualization

When a flight is selected, the route between the origin and destination airports is highlighted on the map.

This interaction helps users quickly understand where the flight is coming from and where it is headed. Route visualization is an important feature because it provides geographical context and improves overall situational awareness.

I also planned the design so that the selected flight becomes the primary focus, making it easier for users to analyze flight-specific information.

User Experience Considerations

Throughout the design process, I focused on keeping the interface clean and easy to navigate.

A dark-themed dashboard was chosen because it is commonly used in monitoring and operations systems. It helps reduce visual strain during long periods of use and creates better contrast for maps, indicators, and operational data.

Consistent spacing, typography, and card layouts were used to improve readability and maintain a professional appearance. The design avoids clutter and presents information in a logical order, helping users find information quickly.

Responsive Design

Although the primary design was created for desktop users, the layout was also planned with responsiveness in mind.

The dashboard structure allows sections such as KPI cards, filters, and flight details to adjust naturally on tablet-sized screens. This ensures that the application remains usable across different devices while maintaining a consistent user experience.

Conclusion

This project allowed me to combine both Angular development and UI/UX design principles to create a practical aviation operations dashboard.

The final solution provides users with an organized way to monitor flights, view operational metrics, visualize routes, and access detailed flight information. My focus throughout the project was on usability, clarity, and creating an interface that supports efficient decision-making in an aviation operations environment.